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School of Vocational Sciences

Kelompok 15
Cybersecurity

SOC LAB: SMART AUTO DOSING PROFILEKEAMANAN INFRASTRUKTUR IOT MELALUI SIMULASI SERANGAN, DETEKSI INSIDEN, DAN MITIGASI BERBASIS SECURITY OPERATIONS CENTER

by: Kelompok 15

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MEET OUR TEAM

KONTEKS PROYEK IOT

Smart Auto Dosing adalah sistem dosing otomatis berbasis IoT. Sensor membaca kondisi, mikrokontroler menggerakkan pompa dosing, lalu data dikirim ke server backend dan ditampilkan di dashboard web.

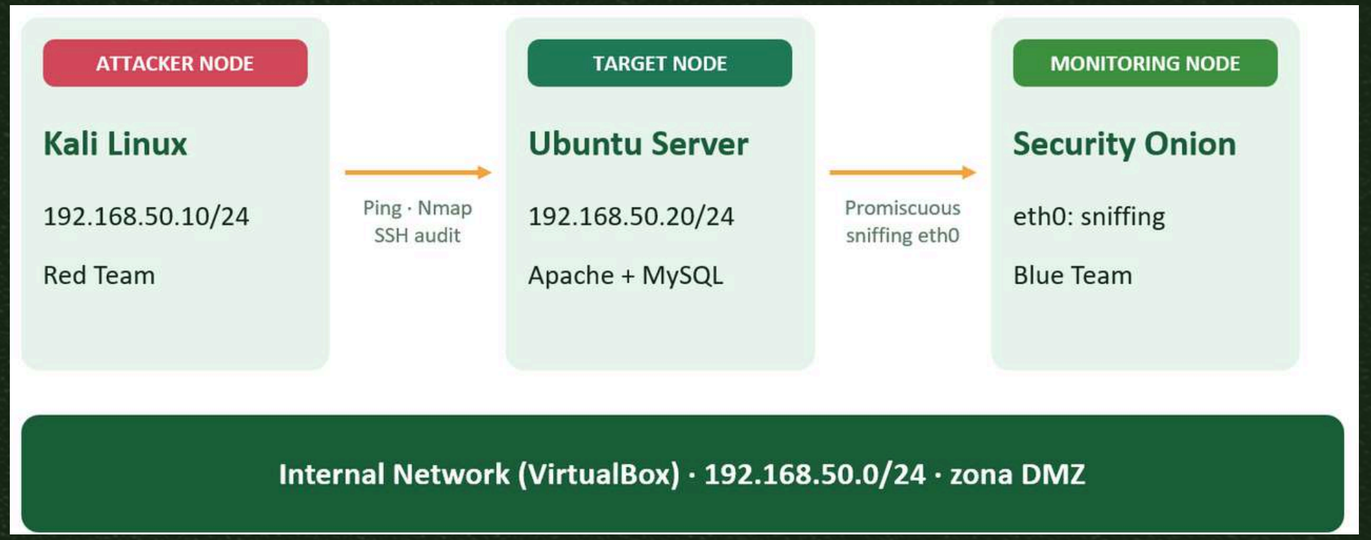
Tujuan Sistem:
Dosing otomatis yang terukur, dengan data dan status perangkat dipantau lewat dashboard web.

Scope tim SOC:
Kami mengamankan server backend replika: SSH, firewall, web Apache, dan database MySQL.

DEKLARASI TECH & REPLIKASI

Komponen	Teknologi pada Proyek IoT Asli	Direplikasi sebagai Target Node
Sistem operasi backend	Linux server	Ubuntu Server VM (192.168.50.20) di VirtualBox
Web server / dashboard	Apache + halaman Smart Auto Dosing	Apache, halaman Smart Auto Dosing | SOC Lab (port 80)
Database / penyimpanan data	MySQL	MySQL lokal, user app_user hak terbatas (data dummy)
Akses administratif	SSH	SSH port 2222 + UFW + fail2ban

ARSITEKTUR SIMULASI



IP PLAN

Node	IP Address	OS	Peran
Target Server	192.168.50.20	Ubuntu Server	Replika backend
Attacker Node	192.168.50.10	Kali Linux	Red Team
Monitoring Node	eth0 tanpa IP (sniffing)	Security Onion	Blue Team

PROGRESS HARDENING & BASELINE

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A. Access & Authentication

ID	Kontrol	Status	Bukti / catatan
A.1	Nonaktifkan login root melalui SSH	Selesai	Hasil `grep PermitRootLogin` → no
A.2	Disable password authentication	Selesai	Verifikasi runtime `sshd -T | grep passwordauthentication` → no
A.3	Ganti port SSH + Fail2ban	Selesai	Nmap diff (22→2222), fail2ban aktif (maxretry 5, bantime 600s)

B. Service & attack surface

ID	Kontrol	Status	Bukti / catatan
B.1	Nonaktifkan service tidak digunakan	Selesai	ModemManager & fwupd dinonaktifkan; service berjalan turun dari 22 → 20
B.2	Firewall ufw whitelist port esensial	Selesai	Port 22 dihapus, 2222 & 80 allow; dikonfirmasi lewat nmap diff
B.3	Update OS	Belum	`apt upgrade` gagal, enp0s3 (dmz) tidak ada akses internet

C. Data & application

ID	Kontrol	Status	Bukti / catatan
C.1	Kredensial tidak hardcoded	Belum	Belum diperiksa pada source code / config aplikasi
C.2	Least-privilege akun database	Selesai	`SHOW GRANTS` → app_user hanya SELECT+INSERT pada iot_dummy.*
C.3	Logging aktivitas mencurigakan	Selesai	`tail /var/log/auth.log` — mencatat percobaan login gagal

PROGRESS HARDENING & BASELINE


```
adminlab@target-server: ~
+------------------------------------------------------------+
adminlab@target-server:~$ sudo sshd -T | grep -E 'permitrootlogin|passwordauthentication|pubkeyauthentication|maxauthtri
es|logingracetime|allowusers'
[sudo] password for adminlab:
logingracetime 120
maxauthtries 6
permitrootlogin no
pubkeyauthentication yes
passwordauthentication no
```

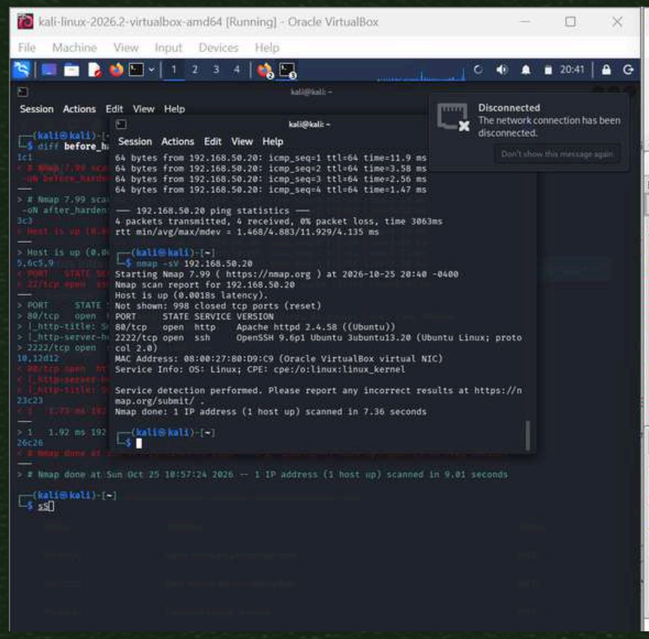
```
adminlab@target-server: ~
passwordauthentication no
adminlab@target-server:~$ sudo ufw status numbered
Status: active

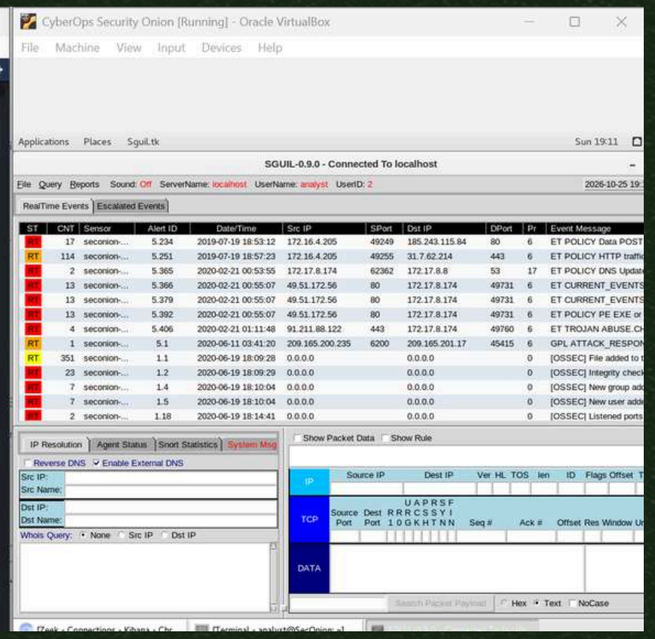
     To                         Action      From
     --                         ------      ----
[ 1] 80/tcp                     ALLOW IN    Anywhere
[ 2] 80/tcp on enp0s8           ALLOW IN    192.168.50.0/24
[ 3] Anywhere                   ALLOW IN    192.168.50.0/24
[ 4] 2222/tcp                   ALLOW IN    Anywhere
[ 5] 80/tcp (v6)                ALLOW IN    Anywhere (v6)
[ 6] 2222/tcp (v6)              ALLOW IN    Anywhere (v6)
```

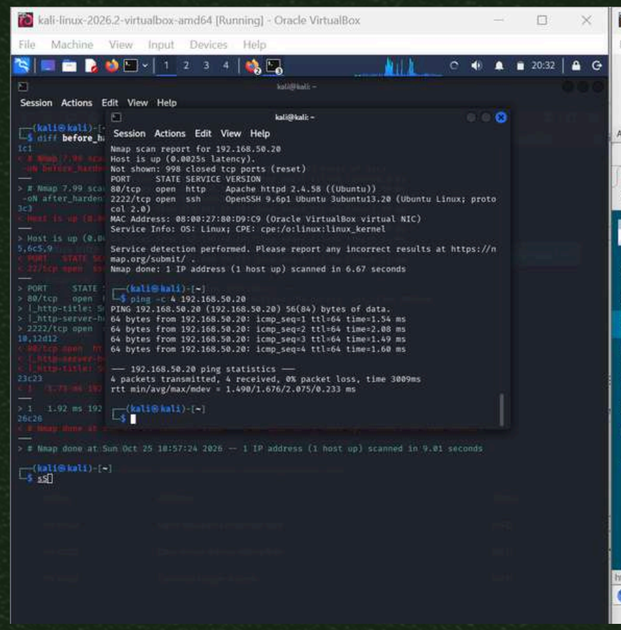
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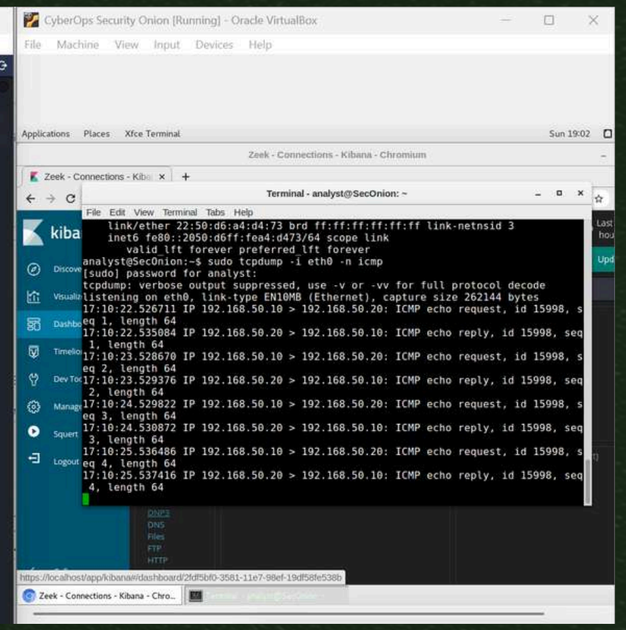
LIVE DEMO

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ATTACK PLAN FASE 2

Target: Ubuntu 192.168.50.20 (replika Smart Auto Dosing) · Attacker: Kali 192.168.50.10

Kategori	Skenario serangan	Tools	Target spesifik
Reconnaissance	Port & service scanning, OS fingerprinting	Nmap	192.168.50.20: port 2222 (SSH) & 80 (HTTP Apache)
Exploitation / akses	Brute-force login SSH	Hydra	Port 2222/SSH, akun adminlab, menguji efektivitas Fail2ban
Web / API / IoT	Enumerasi & akses tidak sah ke web dashboard	Burp Suite, curl	http://192.168.50.20: dashboard Apache & struktur direktori

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KENDALA TEKNIS & PEMBELAJARAN

Kendala teknis	Pembelajaran utama
1. Target Node (Ubuntu) kehilangan akses internet setelah adapter diset ke Internal Network (dmz), sehingga instalasi paket sempat terkendala.	1. Memahami perbedaan NAT, Internal Network, dan Host-only pada VirtualBox serta kapan masing-masing digunakan.
2. Rule firewall (ufw) lama untuk port SSH default tidak otomatis hilang saat port diganti, sehingga perlu dihapus manual.	2. Memahami bahwa perubahan konfigurasi service (sshd_config) tidak otomatis menyesuaikan rule firewall; keduanya harus dikelola terpisah.
3. Traffic berhasil ditangkap di level jaringan (tcpdump) pada interface sniffing Security Onion, namun belum sepenuhnya tervisualisasi di dashboard Kibana.	3. Memahami alur pipeline monitoring SOC (capture, Zeek/Suricata, Elasticsearch, Kibana) dan pentingnya verifikasi tiap tahap secara terpisah.


Ensure strong Leadership
Assemble a Crisis Management
Develop a Crisis Management Plan
Social media communication

THANK YOU